

Network Devices & Infrastructure Report

Comprehensive Analysis: Home Setups, Device Port Schematics, Application Routing, and Campus Troubleshooting

1. Home WiFi Setup: Main Functions & Typical Ports

In a standard home network architecture, various hardware devices collaborate to connect local clients to the Internet. The table below outlines their core operational functions and standard physical ports.

Device	Main Functions	Typical Ports
Switch	Connects devices within a local area network (LAN) using packet switching. Forwards data packets based on MAC addresses, creating dedicated collision domains per port.	Multiple RJ45 Ethernet ports (Fast/Gigabit Ethernet), Uplink port, Power input.
Router	Connects multiple networks together (LAN to WAN/Internet). Performs routing based on IP addresses, NAT (Network Address Translation), DHCP IP allocation, and basic packet filtering.	1x WAN port (RJ45), 4x LAN switch ports (RJ45), Power DC jack, Optional USB port.
Hub	A legacy Layer 1 broadcast device. Repeats incoming electrical signals across all other ports without filtering or MAC address awareness, leading to high collision rates.	Multiple RJ45 ports, Power input (no intelligent management ports).
Firewall	Monitors and filters incoming and outgoing network traffic based on predefined security rules. Establishes a security barrier between trusted internal networks and untrusted external networks.	WAN/External Ethernet ports, LAN/Internal Ethernet ports, DMZ port, Console port (RJ45/Serial), Power input.
Access Point (AP)	Acts as a bridge between wireless Wi-Fi clients and the wired Ethernet network, translating radio frequency (RF) signals into Ethernet data frames.	1x RJ45 Ethernet port (often PoE - Power over Ethernet), Power adapter jack, Reset button.

2. Device Identification & Port Schematics

Below are structural breakdowns and port labels for three core networking devices frequently deployed in modern topologies:

Device A: Wireless Router (Home Gateway)

Primary Identification: Combines routing, switching, and wireless access point capabilities into a single enclosure.

- Marked Ports:**
- WAN Port (Ethernet RJ45)
 - LAN Switch Ports (4x RJ45)
 - Power Jack (DC 12V)
 - External Antennas (RP-SMA / Fixed)
 - Reset / WPS Button

Device B: Network Switch (Managed / Unmanaged)

Primary Identification: Flat box with numerous high-density RJ45 ports and LED indicators for link/activity status on each port.

- Marked Ports:**
- 8x / 24x Gigabit Ethernet Ports (RJ45)
 - Console Port (RJ45-to-Serial / USB)
 - Power Input (AC Inlet / DC Jack)
 - SFP Uplink Slots (Optional Fiber)

Device C: Wireless Access Point (Ceiling / Wall Mount)

Primary Identification: Sleek low-profile disk or box designed for broad radio frequency coverage with minimal physical connectors.

Marked Ports: Ethernet Port with PoE (Power over Ethernet) Console Port (Micro-USB / Pinhole)
Power DC Jack (Alternative to PoE) Internal / External Antenna Array

3. Role of Network Devices in Daily Apps (Zomato)

Application: Zomato (Food Delivery & Restaurant Discovery)

When searching for restaurants or placing an order on Zomato, two primary network devices orchestrate the data flow:

- 1. Wireless Access Point (AP) / Router:** Captures high-frequency radio waves from your smartphone when you tap to load restaurant menus, converting Wi-Fi frames into wired electrical Ethernet packets sent across the local network.
- 2. Firewall / Router:** Intercepts your encrypted HTTPS request, performs Network Address Translation (NAT) to map your private local IP to a public IP, and enforces stateful packet inspection to ensure secure communication with Zomato's cloud backend servers.

4. College WiFi Troubleshooting & Capacity Upgrade

Scenario Analysis & Architectural Recommendations

Identified Bottleneck / Root Cause: College campuses experience severe density challenges where hundreds of students connect to legacy Access Points (e.g., Wi-Fi 4 / 802.11n or early Wi-Fi 5). These older APs have limited concurrent client associations, narrow channel widths (leading to co-channel interference), and lack advanced multi-user scheduling (MU-MIMO/OFDMA). Consequently, when lecture halls fill up, airtime contention skyrockets, causing high packet loss, latency spikes, and frequent disconnections.

Proposed Upgrade Solution: Upgrading the campus Access Points to enterprise-grade **Wi-Fi 6 or Wi-Fi 6E APs**—paired with multi-gigabit PoE+ switches—drastically improves capacity and traffic handling. Wi-Fi 6 introduces OFDMA and uplink/downlink MU-MIMO, allowing the AP to communicate with multiple client devices simultaneously rather than sequentially. This eliminates airtime congestion, supports high-density concurrent connections, and restores stable, high-speed campus connectivity.